

Taehoon Yoon

Email: taehoon@kaist.ac.kr | Phone: +82-10-4787-2725 | GitHub: [taehoon-yoon](https://github.com/taehoon-yoon)

ACADEMIC INTERESTS

Generative Modeling, Probabilistic Inference: Diffusion and Flow-Matching Models, Controllable Generation, Reward Alignment, Inverse Problems, Particle-Based Sampling Methods

EDUCATION

University of Michigan, Ann Arbor

2026 (expected) -

• Ph.D. in Electrical & Computer Engineering

Advisor: [Liyue Shen](#)

Korea Advanced Institute of Science and Technology (KAIST)

2024 - 2026

• Master of Science in Artificial Intelligence

Advisor: [Minhyuk Sung](#)

Sogang University

2017 - 2024

• Bachelor of Science in Physics

• Bachelor of Science in Computer Science and Engineering

Graduated as the **top student** from the College of Natural Sciences (**Valedictorian**)

GPA: 4.27 / 4.3

PUBLICATIONS

* denotes equal contribution. C: Conference paper. W: Workshop paper. P: Preprint.

[W1] Contrastive Distribution Matching for Amortized Sequential Monte Carlo in Discrete Diffusion

Jaihoon Kim, **Taehoon Yoon**, Prin Phunyaphibarn, Seungjun Kim, Morteza Mardani, Minhyuk Sung

ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling

[Website](#) [Paper](#) [arXiv](#) [Code](#)

[C3] Ψ -Sampler: Initial Particle Sampling for SMC-Based Inference-Time Reward Alignment in Score Models

Taehoon Yoon*, Yunhong Min*, Kyeongmin Yeo*, Minhyuk Sung

NeurIPS 2025, Spotlight (top 3.2%)

[Website](#) [Paper](#) [arXiv](#) [Code](#)

[C2] Inference-Time Scaling for Flow Models via Stochastic Generation and Rollover Budget Forcing

Jaihoon Kim*, **Taehoon Yoon***, Jisung Hwang*, Minhyuk Sung

NeurIPS 2025

[Website](#) [Paper](#) [arXiv](#) [Code](#)

[C1] GrounDiT: Grounding Diffusion Transformers via Noisy Patch Transplantation

Phillip Y. Lee*, **Taehoon Yoon***, Minhyuk Sung

NeurIPS 2024

[Website](#) [Paper](#) [arXiv](#) [Code](#)

HONORS AND AWARDS

- **Valedictorian**, College of Natural Sciences, Sogang University
 - **Dean's List**, Sogang University
 - **Top 1% Grade Award**, Sogang University
- Feb. 2024

Feb. 2023, Feb. 2020

Sept. 2020, Mar. 2020, Mar. 2019

RESEARCH EXPERIENCE

- Visual AI Lab, KAIST*
Graduate Researcher

Visual AI Lab, KAIST
Research Intern

CJ AI Center, CJ Group
Research Intern (hosted by Seoul National University AI Institute)

Visual Computing Lab, Sogang University
Undergraduate Research Assistant

BISPL Lab, KAIST
Research Intern

Mar. 2024 - Dec. 2025

Jan. 2024 - Feb. 2024

Jul. 2023 - Aug. 2023

Sept. 2022 - Apr. 2023

Jul. 2022 - Sept. 2022

Advisor: [Minhyuk Sung](#)

Advisor: [Minhyuk Sung](#)

Advisor: [Seong Jong Ha](#)

Advisor: [Joo Ho Lee](#)

Advisor: [Jong Chul Ye](#)

WORK EXPERIENCE

- Donga Science Observatory*
Observatory & Education Assistant

May. 2018 - Feb. 2019

ACADEMIC SERVICE

- Reviewer**
NeurIPS

2026